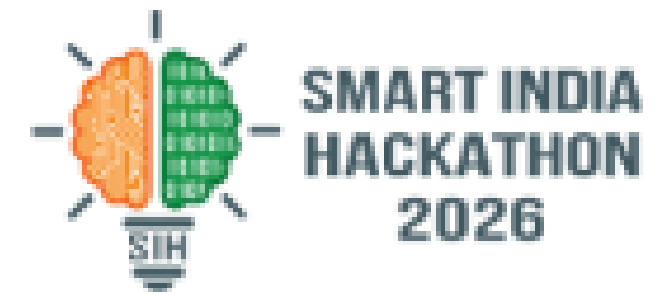
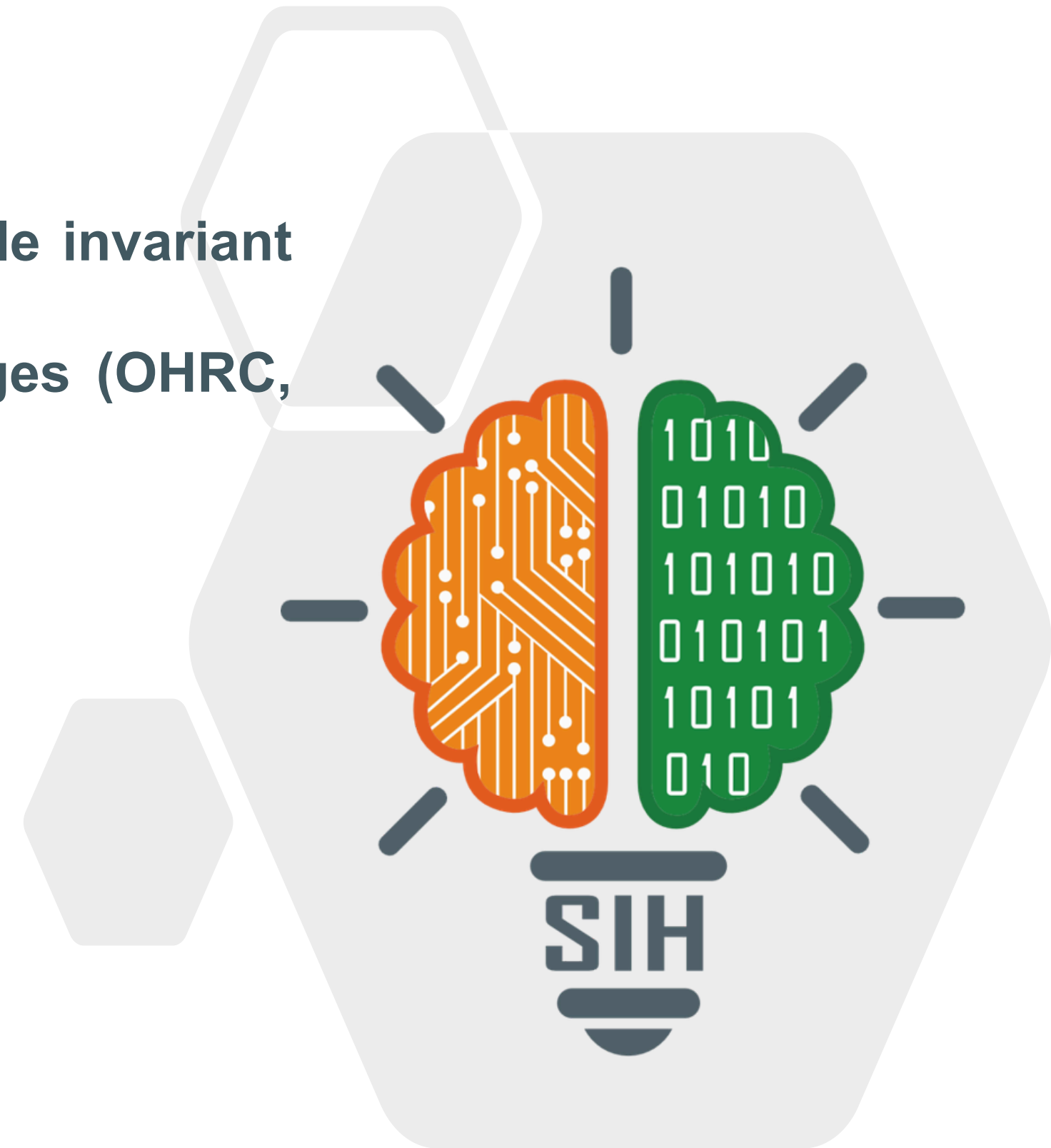


SMART INDIA HACKATHON 2026



- Problem Statement ID - SIH26166
- Problem Statement - Title-Multi-modal, Sun angle and scale invariant image correspondence using Chandrayaan-2 optical images (OHRC, TMC and IIRS)
- Theme - Space Technology
- PS Category - Software
- Team ID -
- Team Name - LunaX



Current Problem

- **Illumination Variation:** Changes in Sun azimuth and elevation affect lunar surface lighting.
- **Viewpoint Variation:** Different camera positions and orientations cause shifts, rotation and perspective distortions.
- **Scale Variation:** Different imaging altitudes and resolutions create large scale differences.
- **Difficult Correspondence:** Finding reliable matching points between images captured under different conditions is challenging.
- **Non-Uniform Matches:** Match points may cluster in limited areas instead of being uniformly distributed across the image.

Proposed Solution

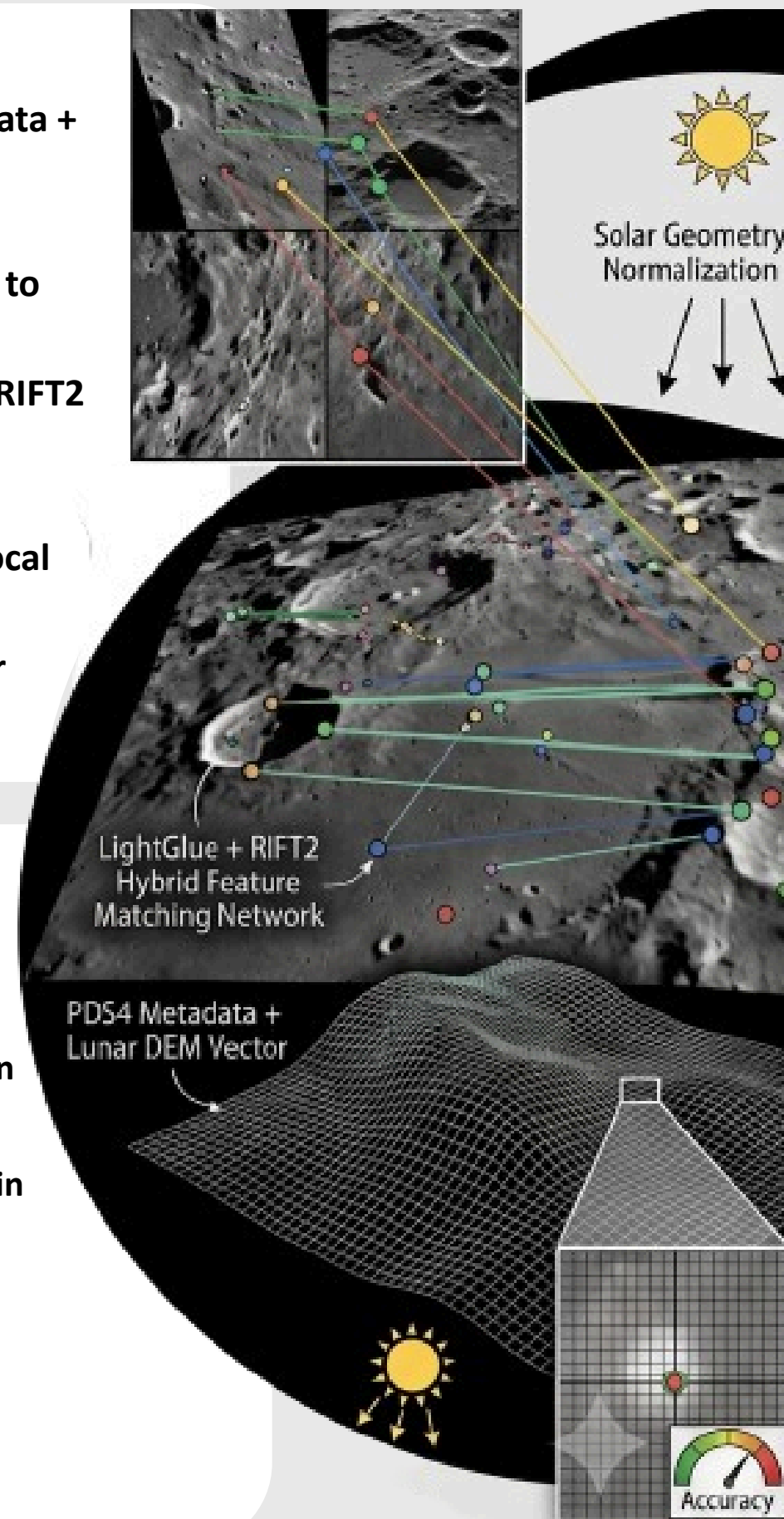
- **Physics-Informed Registration:** Uses PDS4 metadata + lunar DEMs to understand camera geometry and terrain.
- **Illumination Normalization:** Uses solar geometry to reduce Sun-angle and lighting variations.
- **Hybrid Feature Matching:** Combines LightGlue + RIFT2 for reliable cross-modal correspondences.
- **Robust Alignment:** Uses spatial bucketing + MAGSAC++ + TPS to remove errors and correct local distortions.
- **Sub-Pixel Refinement:** Uses Phase Correlation for precise source-to-reference alignment.

Key Features

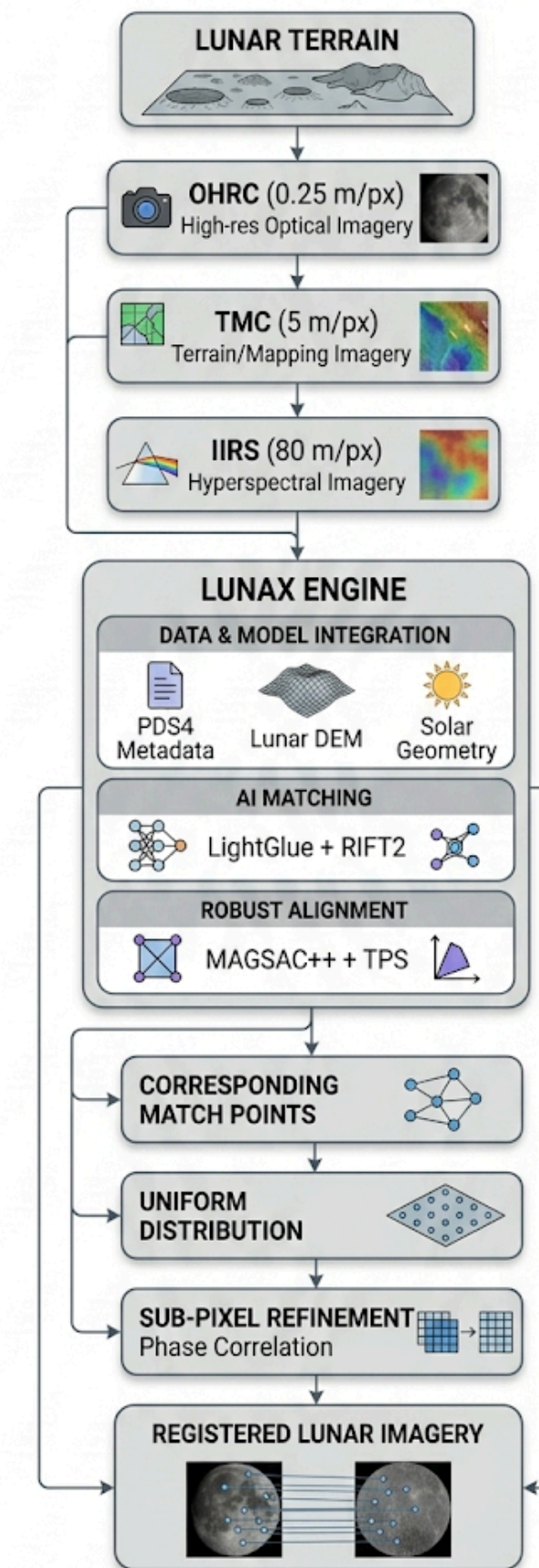
- **Multi-Sensor Correspondence:** Matches features across OHRC, TMC and IIRS imagery.
- **Scale & Viewpoint Robustness:** Handles differences in resolution, altitude and camera geometry.
- **Illumination-Aware Matching:** Accounts for changes in Sun position, shadows and surface brightness.
- **Uniform Match Distribution:** Ensures reliable correspondences are spread across the image.
- **Sub-Pixel Accuracy:** Refines alignment beyond pixel-level precision for highly accurate registration.

Why we stand out?

- **Scientific Advancement:** Enables richer lunar analysis by combining multi-sensor data.
- **Future Exploration:** Supports precise lunar mapping and future mission planning.
- **High-Precision Mapping:** Delivers uniform match points and sub-pixel alignment for accurate analysis.



LunaX



Tech Stack

Programming

- python 3.10+

Computer Vision & Image Processing

- OpenCV
- Scikit-image
- Kornia

Deep Learning

- PyTorch
- SuperPoint
- LightGlue
- RIFT2

Interface

- Typer / Click
- Streamlit /Gradio

Geospatial Processing

- GDAL
- Rasterio
- PyProj

Scientific Computing

- Numpy
- SciPy

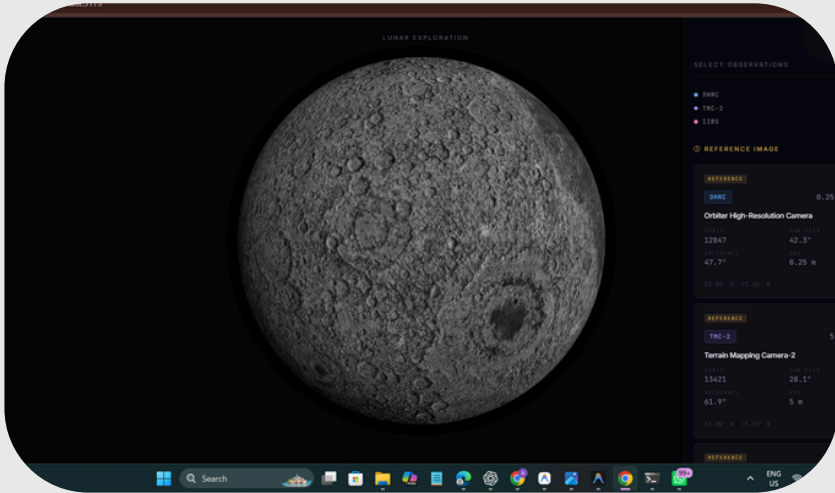
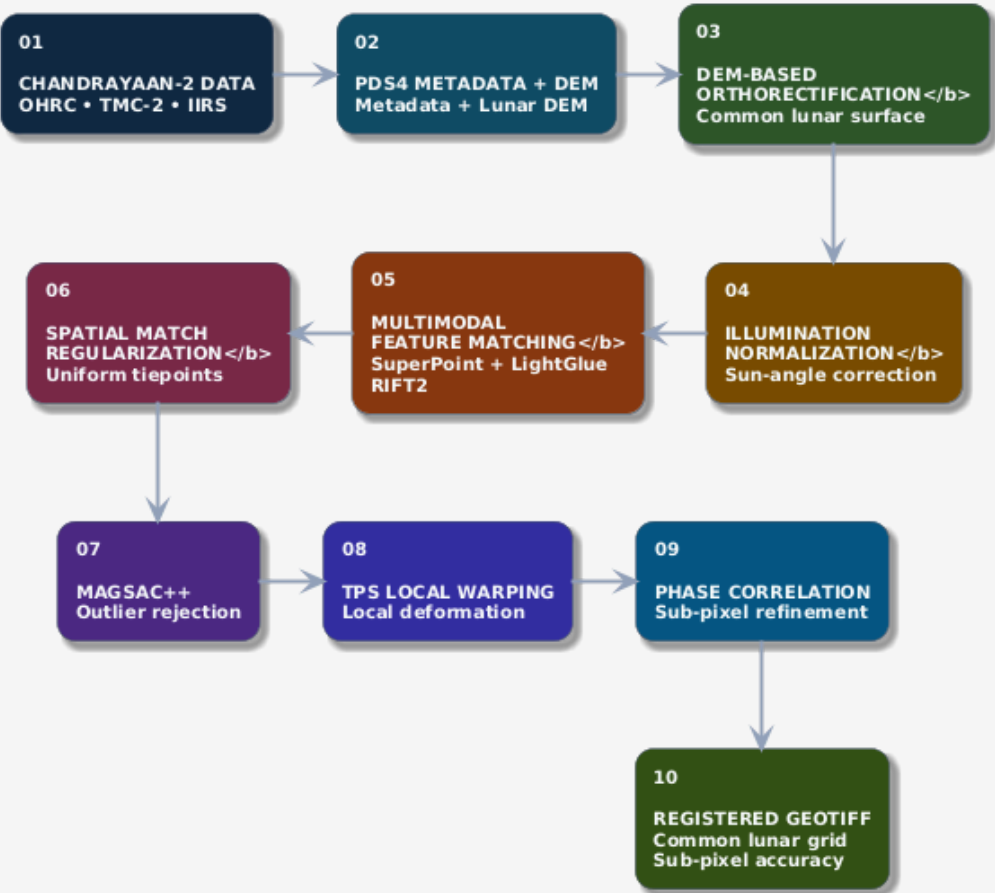
Lunar Data & Ingestion

- pds4_tools
- PDS4 / GeoTIFF

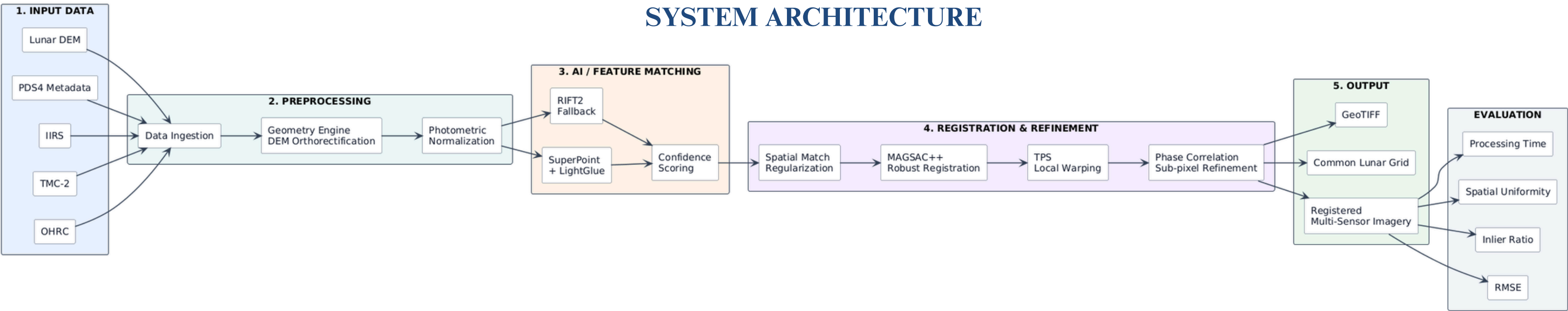
Visualization & Evaluation

- Matplotlib
- SeaBorn

REGISTRATION PIPELINE



SYSTEM ARCHITECTURE



Feasibility Analysis

- Technical
- Feasible: PDS4 provides camera, spacecraft and Sun-angle data.
 - Challenge: Large scale and modality differences affect matching.
 - Solution: DEM + illumination correction + LightGlue/RIFT2.
- Operational
- Feasible: Automated multi-stage processing pipeline.
 - Challenge: Large images require high computation.
 - Solution: Efficient matching with tiled processing.
- Financial
- Low-Cost: Uses existing datasets and open-source methods.
 - Challenge: Large-scale processing may require GPUs.
 - Solution: Optimize computation and use scalable resources.
- Evaluation
- RMSE: Measures the positional error of matched points.
 - Inlier Ratio: Measures the percentage of reliable matches.
 - Match Uniformity: Checks how evenly match points are distributed across the image.

Feasibility Analysis

TECHNICAL	OPERATIONAL	FINANCIAL	EVALUATION
			
STRENGTHS Physics-informed registration	STRENGTHS Automated registration pipeline	STRENGTHS Open datasets & software methods	STRENGTHS Measurable registration quality
CHALLENGES Scale, viewpoint & illumination variations	CHALLENGES Large lunar image processing	CHALLENGES GPU computation requirements	CHALLENGES Need accurate correspondences
STRATEGIES DEM + Solar correction + AI matching	STRATEGIES Tiled & efficient processing	STRATEGIES Optimized & scalable processing	STRATEGIES RMSE + Inlier Ratio + Match Uniformity

SWOT Analysis

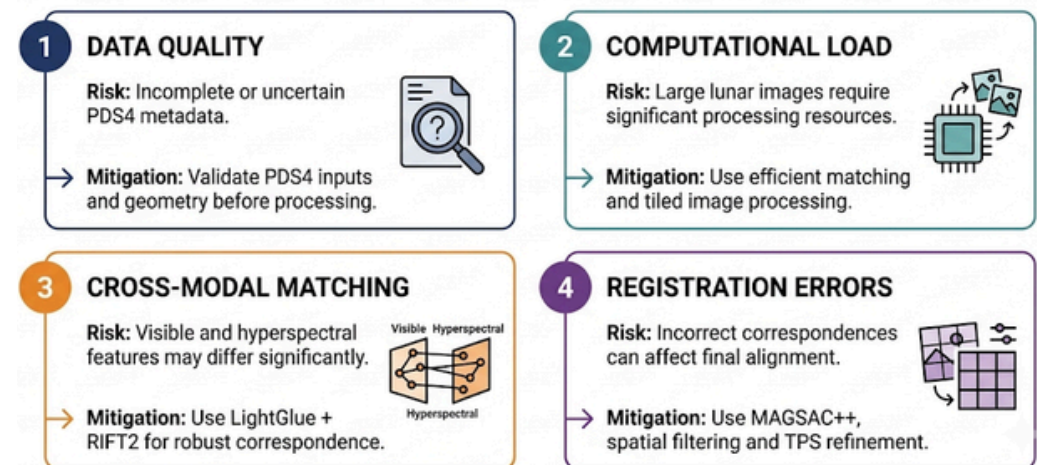


Viability Analysis

- **Technical Viability:** Combines DEM projection, photometric correction, feature matching and robust fitting into one workflow.
- **Operational Viability:** Follows an automated flow from PDS4 input → registration → correspondence → final product.
- **Scientific Viability:** Brings Chandrayaan-2 sensor data into a common geometry for multi-sensor lunar analysis.
- **Performance Viability:** Produces registered images and match points, evaluated using RMSE and other metrics.

Challenges & Strategies

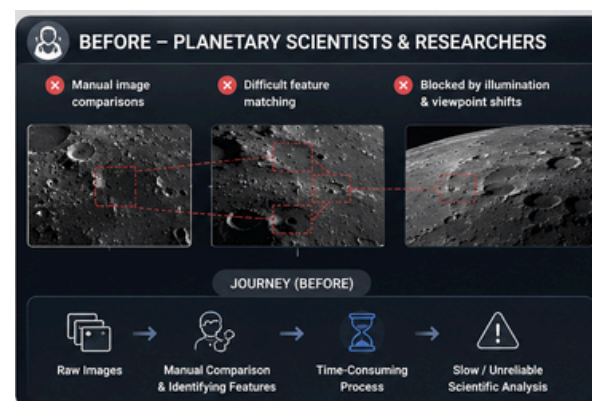
- Technical
- Challenge: Large scale and sensor differences make feature matching difficult.
 - Strategy: Use LightGlue + RIFT2 for robust correspondence.
- Geometric
- Challenge: Lunar terrain causes viewpoint and perspective distortions.
 - Strategy: Use DEM-based projection + TPS correction.
- Illumination
- Challenge: Different Sun angles change shadows and surface brightness.
 - Strategy: Apply solar geometry normalization.
- Correspondence
- Challenge: Matches may cluster in limited regions.
 - Strategy: Use spatial bucketing for uniform distribution.
- Precision
- Challenge: Pixel-level alignment may not be sufficient.
 - Strategy: Use Phase Correlation for sub-pixel refinement.



Impact on Different Users & Their Journey

PLANETARY SCIENTISTS & RESEARCHERS

- **BEFORE:** Manual image comparisons; difficult feature matching; blocked by illumination & viewpoint shifts.
- **AFTER:** Auto-registered imagery; reliable cross-sensor correspondence; accelerated analysis.
- **JOURNEY:** Raw Images → Registration → Corresponding Features → Scientific Analysis



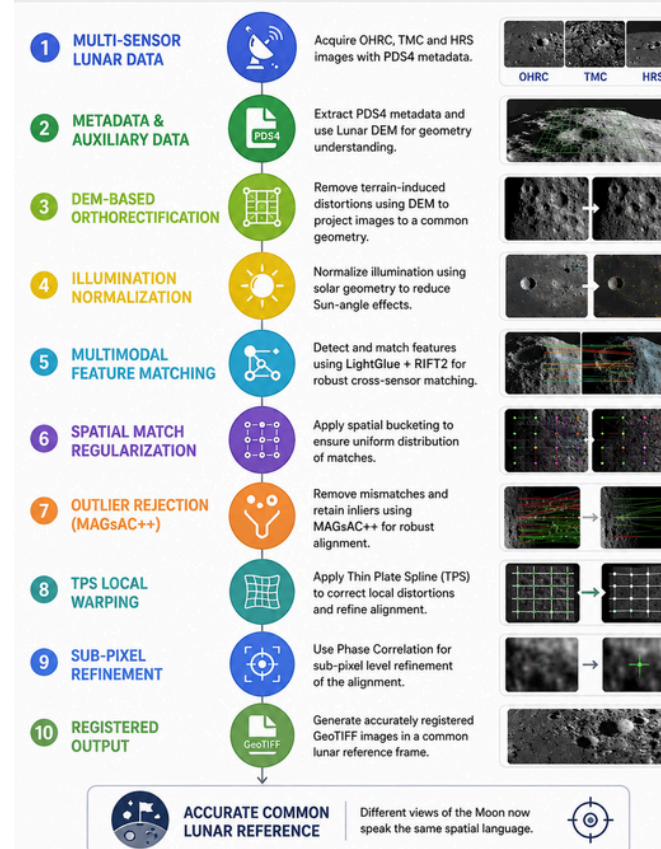
MISSION PLANNERS & SPACE AGENCIES

- **BEFORE:** Fragmented sensor datasets; complex mapping due to varying resolutions; time-heavy manual prep.
- **AFTER:** Unified datasets; rapid mission-ready maps; superior landing-site analysis.
- **JOURNEY:** Multi-Sensor Data → Spatial Reference → Mission-Ready Maps → Better Planning



SCIENTIFIC, TECHNICAL & OPERATIONAL IMPACT

- **SCIENTIFIC:** Enables precise OHRC/TMC/HRS comparisons for advanced geological and temporal lunar studies
- **TECHNICAL:** Achieves robust, sub-pixel matching under varying conditions, outputting standard GeoTIFFs.
- **OPERATIONAL:** Fully automates alignment via a scalable, reusable methodology.



IMPACT ON SPACE AGENCIES, RESEARCHERS & FUTURE MISSIONS



Alignment with UN SDGs

- **SDG 4 (Quality Education):** Empowers STEM and academic research with unified lunar datasets.
- **SDG 9 (Innovation & Infrastructure):** Advances space tech infrastructure using AI image processing.
- **SDG 10 (Reduced Inequalities):** Democratizes access to high-precision space data for global researchers.
- **SDG 12 (Responsible Consumption):** Maximizes Chandrayaan-2 data utility to reduce redundant missions.
- **SDG 16 (Strong Institutions):** Ensures transparent and reproducible planetary mapping workflows.
- **SDG 17 (Partnerships):** Leverages PDS4 standards for seamless international space collaboration.

References

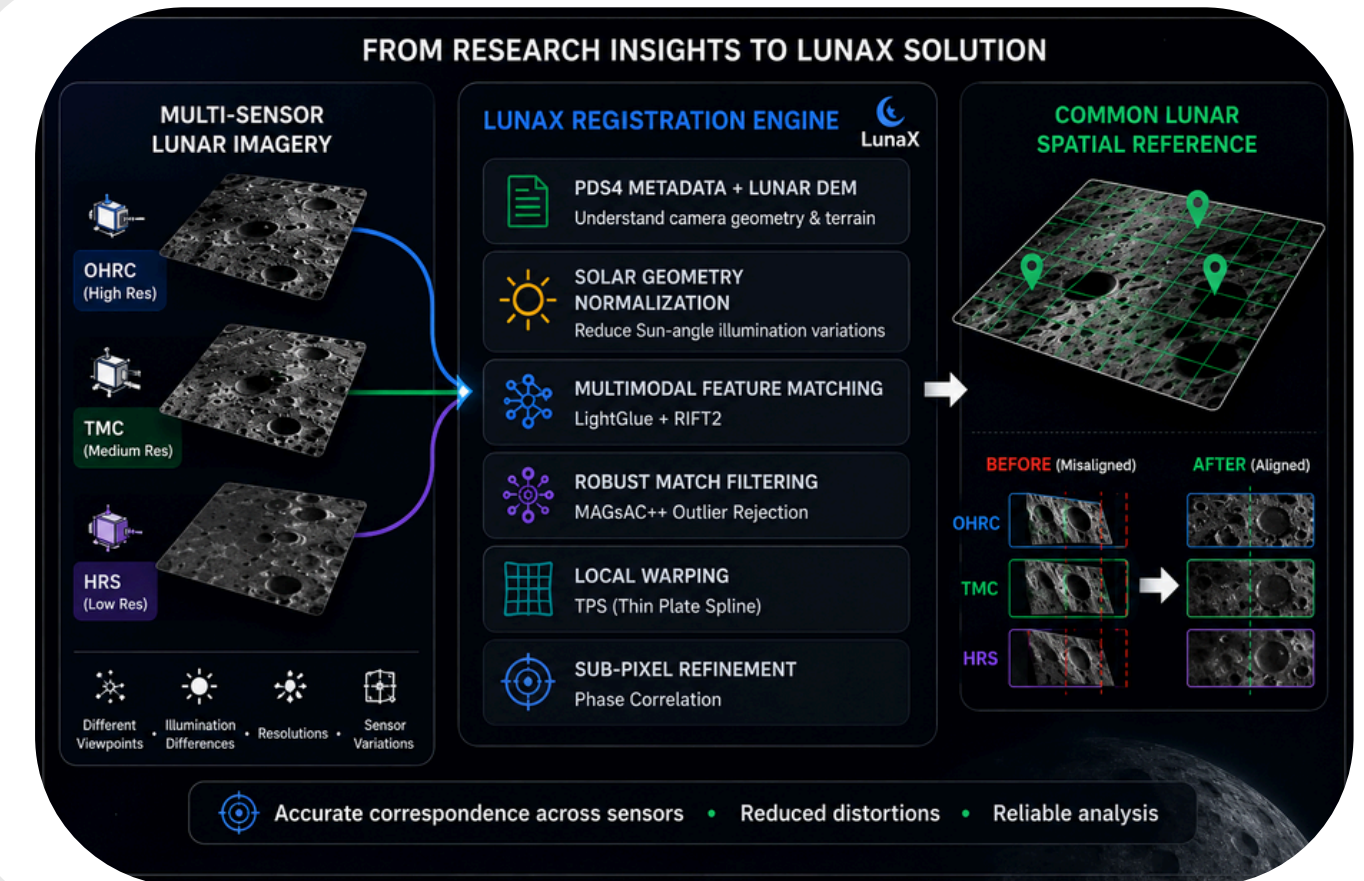
- ISRO / Chandrayaan-2 — OHRC, TMC & IIRS instrument and mission data. [link](#)
- Makharia et al., 2025 — Chandrayaan-2 Feature Matching Benchmark ISRO-SAC [link](#)
- Kumar et al., 2024 — MoonMetaSync: Lunar Image Registration Analysis IEEE WNYISPW 2024. [link](#)
- Sarlin et al., 2020 — SuperGlue: Deep Feature Matching CVPR 2020. [link](#)
- Lindenberger et al., 2023 — LightGlue: Fast Feature Matching ICCV 2023. [link](#)

Key insights from research

- **Multi-Sensor Registration:** OHRC, TMC and IIRS provide complementary observations but differ significantly in resolution and imaging characteristics.
- **Geometry Matters:** Lunar terrain, camera viewpoint and acquisition geometry can introduce significant registration errors.
- **Illumination Matters:** Different Sun azimuth and elevation change shadows and surface appearance, making direct feature matching difficult.
- **Multimodal Matching:** Conventional intensity-based methods may struggle when images from different sensors represent the same terrain differently.
- **Robust Correspondence:** Reliable registration requires outlier rejection and well-distributed match points across the image.

Demo Video Link
[CLICK ME](#)

Prototype Link
[CLICK ME](#)



Key Realisations

- **Physics Before Matching:** Correct geometry and illumination variations before performing feature correspondence.
- **Hybrid Matching:** Combine complementary feature-matching approaches to improve robustness across different image modalities.
- **Uniformity Matters:** Matching accuracy should not depend on features concentrated in only one region.
- **Precision is Essential:** Final alignment requires refinement beyond ordinary pixel-level registration.
- **Measurable Output:** The system should produce registered imagery + corresponding match points, evaluated using RMSE, inlier ratio and match distribution.